

Solved Example Problem with Stokes Theorem

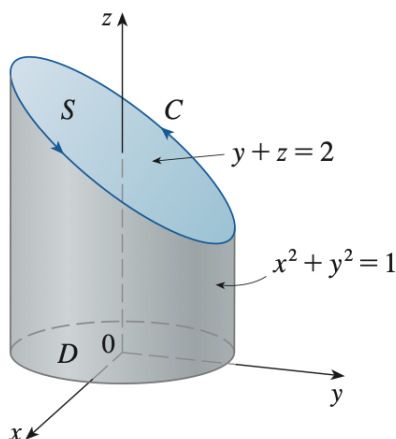
S25 Math 164: Multivariable Calculus

Leo Liang

This is a full in-detail solution to example 1 in Chapter 16.8 (Stokes' Theorem) of the class textbook, *Calculus Early Transcendental 9th Edition*. I found that the textbook lacked a lot of intermediary steps which may make it hard for students to follow.

Problem

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y, z) = -y^2\mathbf{i} + x\mathbf{j} + z^2\mathbf{k}$ and C is the curve of intersection of the plane $y + z = 2$ and the cylinder $x^2 + y^2 = 1$. Orient C to be counterclockwise when viewed from above. Here is a picture of C ,



Solution

We use fact of Stokes Theorem

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$$

and choose the elliptical region S in the plane $y + z = 2$, which is bounded by C , as show in the picture. Note that we could've picked any surface with C as its boundary, but the elliptical region is the simplest one and will be most convenient to use. We first find $\text{curl } \mathbf{F}$

$$\text{curl } \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y^2 & x & z^2 \end{vmatrix} = (1 + 2y)\mathbf{k} = \langle 0, 0, 1 + 2y \rangle.$$

So at this point we have

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S} = \iint_S \langle 0, 0, 1 + 2y \rangle \cdot d\mathbf{S}.$$

Then we use the equation 9 in Chapter 16.7 (Surface Integrals) which states

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n} \, dS = \iint_D \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v) \, dA$$

Note: the above equation is essential to solving the problem! The textbook did not explicitly mention this when solving the example, as it assumed the reader already knew this. We must parameterize our surface S which is part of the plane $y + z = 2$. We can write $z = f(x, y) = 2 - y$ then use the trivial parameterization

$$x = x \quad y = y \quad z = 2 - y$$

So our parameterized surface is

$$\mathbf{r}(x, y) = \langle x, y, 2 - y \rangle.$$

Note: this is the parameterization for the entire plane. We will restrict the plane to only the elliptical region later when setting up our double integral. Then we have

$$\begin{aligned} \mathbf{r}_x &= \langle 1, 0, 0 \rangle \\ \mathbf{r}_y &= \langle 0, 1, -1 \rangle. \end{aligned}$$

Then

$$\mathbf{r}_x \times \mathbf{r}_y = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 0 \\ 0 & 1 & -1 \end{vmatrix} = \langle 0, 1, 1 \rangle$$

Note: you could've obtained this by observing the normal vector of the plane $y + z = 2$ is $\langle 0, 1, 1 \rangle$ since we can write $0x + 1y + 1z = 2$ and the coefficients are the normal vector. Now piecing everything together we have

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S} \\ &= \iint_S \langle 0, 0, 1 + 2y \rangle \cdot d\mathbf{S} \\ &= \iint_D \langle 0, 0, 1 + 2y \rangle \cdot \langle 0, 1, 1 \rangle \, dA \quad (\text{by equation 9}) \\ &= \iint_D (1 + 2y) \, dA \end{aligned}$$

Note: in the textbook you will see that last equation without showing the intermediary steps. Recall earlier that we parameterized the entire plane $y + z = 2$, but now we can restrict it to our desired elliptical region. Observe that the domain of the parameterization $\mathbf{r}(x, y) = \langle x, y, 2 - y \rangle$ which map to our desired elliptical region S is simply all the points in the circle $x^2 + y^2 = 1$ in the xy -plane, as shown in the picture by region D . (this sentence is a mouthful, but really try to understand what is happening here. Ask yourself, what domain of $\mathbf{r}(x, y) = \langle x, y, 2 - y \rangle$ gives us the elliptical surface S as desired?). Since our domain D is a circle, we convert to polar coordinates for ease of calculation

$$x = r \cos(\theta) \quad y = r \sin(\theta),$$

then

$$\begin{aligned} \int_C \mathbf{F} \cdot d\mathbf{r} &= \iint_D (1 + 2y) \, dA \\ &= \int_0^{2\pi} \int_0^1 (1 + 2r \sin(\theta)) \, dr \, d\theta \\ &= \int_0^{2\pi} \left[\frac{r^2}{2} + 2 \frac{r^3}{3} \sin \theta \right]_0^1 d\theta \\ &= \int_0^{2\pi} \left(\frac{1}{2} + \frac{2}{3} \sin \theta \right) d\theta \\ &= \frac{1}{2}(2\pi) + 0 = \pi \quad \square \end{aligned}$$

Please feel free to email me at lliang11@u.rochester.edu if you have any questions.